SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	**CO4:** Articulation (BL4, BL5)	**Assessment:**	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	20% of CIA	**Total Marks:**	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
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Section / Batch:	CSE	**Date:**	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues.
- Provide a thorough analysis with validated results and performance evaluation.

Question Type	Focus Area	Questions
Industry Problem Solving Task	Focus on Solving optimization problems using dynamic programming and greedy strategies	Q1

SECTION A – Industry Problem Solving Task

Code of Conduct

I **J KARTHIK / 192472136** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: J KARTHIK

Question 1 — Knapsack: Industry Resource Allocation

Marks: 20

A cloud service provider must allocate limited server memory (100 GB) among multiple client applications, each having different memory requirements and revenue values. Analyze the problem and explain how the 0/1 Knapsack algorithm can be applied to maximize total revenue. Design a feasible solution using dynamic programming and justify the selection of this approach. Discuss how overlapping subproblems and optimal substructure are utilized. Suggest suitable tools or platforms for implementation. Analyze the performance in terms of time and space complexity and interpret the results in an industry context.

1. Understanding the Industry Problem

The cloud provider has a fixed resource pool — 100 GB of server memory. A set of n client applications each request a fixed amount of memory (weight, wi) and each, if hosted, generates a fixed revenue (value, vi). Because an application either runs fully on the server or is not deployed at all (a VM/container cannot be "half allocated"), this is a classic 0/1 Knapsack problem: choose a subset of applications whose total memory does not exceed 100 GB while maximizing total revenue.

- Key issue 1 — Indivisibility: applications cannot be partially hosted, ruling out the fractional/greedy knapsack variant.
- Key issue 2 — Hard capacity constraint: total allocated memory must never exceed 100 GB (a hosting SLA breach otherwise).
- Key issue 3 — Combinatorial explosion: a brute-force search of all 2^n subsets is infeasible once the number of client applications grows beyond ~25–30.

2. Why Dynamic Programming (and not Greedy)

A greedy approach (e.g., picking apps by highest revenue-per-GB ratio) fails here because it can lock in a locally good choice that blocks a globally better combination — 0/1 Knapsack does not have the greedy-choice property that fractional knapsack has. Dynamic programming instead guarantees the global optimum by exploiting two properties of the problem:

Optimal substructure: the best allocation for capacity W using the first i applications is built from the best allocation for a smaller capacity/smaller application set — i.e., dp[i][W] depends only on optimal solutions to dp[i-1][*].

Overlapping subproblems: the same sub-instance "best revenue using the first i apps with capacity w" is required repeatedly while evaluating different combinations, so a naive recursive solution recomputes it many times. DP stores each subproblem's answer once in a table and reuses it, converting an exponential recursion tree into a polynomial-time table fill.

3. Solution Design

Let there be n applications with memory[i] and revenue[i], and capacity W = 100. Define:

dp[i][w] = maximum revenue obtainable using the first i applications with a memory budget of w.

Recurrence relation:

dp[0][w] = 0 for all w
dp[i][w] = dp[i-1][w] if memory[i] > w
dp[i][w] = max(dp[i-1][w],
dp[i-1][w-memory[i]] + revenue[i]) otherwise
The final answer is dp[n][W]. Backtracking through the table (comparing dp[i][w] to dp[i-1][w]) recovers which applications were actually selected.

4. Sample Implementation (Python)

```
def knapsack_allocation(memory, revenue, capacity):
    n = len(memory)
    dp = [[0] * (capacity + 1) for _ in range(n + 1)]

    for i in range(1, n + 1):
        for w in range(capacity + 1):
            if memory[i - 1] > w:
                dp[i][w] = dp[i - 1][w]
            else:
                dp[i][w] = max(dp[i - 1][w],
                               dp[i - 1][w - memory[i - 1]] + revenue[i - 1])

    # Backtrack to find selected applications
    w, selected = capacity, []
    for i in range(n, 0, -1):
        if dp[i][w] != dp[i - 1][w]:
            selected.append(i - 1)
            w -= memory[i - 1]

    return dp[n][capacity], selected[::-1]


apps     = ['App-A', 'App-B', 'App-C', 'App-D', 'App-E']
memory   = [10, 20, 30, 40, 50]      # GB required by each app
revenue  = [60, 100, 120, 160, 200]  # revenue generated (USD/day)
capacity = 100                       # total server memory (GB)
```

```
max_revenue, chosen = knapsack_allocation(memory, revenue, capacity)
print('Maximum total revenue:', max_revenue)
print('Applications deployed:', [apps[i] for i in chosen])
print('Memory used:', sum(memory[i] for i in chosen), 'GB out of', capacity, 'GB')
```

5. Sample Input and Output

Input: memory = [10, 20, 30, 40, 50] GB, revenue = [60, 100, 120, 160, 200] USD/day, capacity = 100 GB.

Maximum total revenue: 440
Applications deployed: ['App-A', 'App-B', 'App-C', 'App-D']
Memory used: 100 GB out of 100 GB
Validation: App-A+B+C+D consume exactly 10+20+30+40 = 100 GB and yield 60+100+120+160 = 440 USD/day, which is provably optimal for this instance — every other feasible combination (e.g., B+C+E = 100 GB → 420, A+D+E = 100 GB → 420) yields strictly less revenue.

6. Suggested Tools / Platforms

- Python (NumPy / pandas) or Java for prototyping the DP table; C++ for latency-critical production allocators.
- AWS/Azure/GCP autoscaling and resource-manager APIs to feed live memory-request and pricing data into the optimizer.
- Kubernetes with a custom scheduler plugin (bin-packing extender) to enforce the knapsack decision at pod-placement time.
- OR-Tools or PuLP for validating the DP result against an ILP formulation on larger instances.

7. Performance Analysis

Time Complexity: O(n × W) — the table has (n+1) rows and (W+1) columns, each cell computed in O(1). For n = 100 apps and W = 100 GB this is only 10,000 operations, versus O(2^n) ≈ 1.27 × 10^30 for brute force.

Space Complexity: O(n × W) for the full table (needed for backtracking); this can be reduced to O(W) using a rolling 1-D array if only the optimal value (not the selected set) is required.

Industry interpretation: Because capacity W is bounded by physical server memory (a fixed, moderate number), the pseudo-polynomial O(n·W) run time is entirely practical for real-time or near-real-time allocation decisions, letting the cloud provider re-optimize allocation every time a new client request or a churned tenant changes the resource pool, directly maximizing daily revenue per server.

Question 2 — Floyd–Warshall: Network Routing

A telecom company needs to compute shortest paths between all pairs of routers in a network. Analyze how the Floyd–Warshall algorithm can be applied to optimize routing. Design a solution by explaining how intermediate nodes improve shortest path computation. Suggest suitable tools or technologies for implementation. Analyze the time and space complexity of the algorithm. Interpret the results in terms of network efficiency and real-world applicability.

1. Understanding the Industry Problem

A telecom backbone can be modeled as a weighted directed graph where routers are vertices and physical/logical links are edges weighted by latency, hop cost, or bandwidth cost. The operator needs the shortest path between every pair of routers simultaneously (not just from one source), so that any router's routing table can be populated in one pass.

- Key issue 1 — All-pairs requirement: running a single-source algorithm (e.g., Dijkstra) from every node is possible but less elegant when the network is dense and edge weights can be negative (e.g., cost credits).
- Key issue 2 — Correctness with negative edge weights: link-cost incentives can sometimes be modeled as negative weights; Floyd–Warshall handles these correctly (as long as there is no negative cycle), unlike Dijkstra.
- Key issue 3 — Need for a complete distance matrix for fast route lookups without re-computation at query time.

2. Why Floyd–Warshall (Dynamic Programming)

Floyd–Warshall incrementally allows each router k (in turn) to act as an intermediate hop for every pair (i, j). This is a textbook dynamic-programming formulation:

Optimal substructure: the shortest path from i to j using only intermediate nodes {1..k} is either the shortest path using only {1..k-1}, or it passes through k, in which case it is the shortest i→k path plus the shortest k→j path (each restricted to intermediates {1..k-1}).

Overlapping subproblems: the same i→k and k→j distances are reused for many different (i, j) pairs across iterations, so tabulating dist[k][i][j] (compressed to a 2-D matrix updated in place) avoids recomputation.

3. Solution Design — Role of Intermediate Nodes

Let dist[i][j] be initialized to the direct edge weight (or ∞ if no direct link, 0 if i = j). For each candidate intermediate router k = 1..n, and for every pair (i, j), the algorithm checks whether routing traffic i → k → j is cheaper than the current best-known i → j path:

```
for k in 1..n:
   for i in 1..n:
      for j in 1..n:
```

```
            if dist[i][k] + dist[k][j] < dist[i][j]:
                dist[i][j] = dist[i][k] + dist[k][j]
```

After router k has been considered as an intermediate hop for all pairs, dist[i][j] is guaranteed optimal over all paths that use only routers {1..k} as intermediates. Once k has swept through all n routers, dist[i][j] holds the true shortest path cost for every pair.

4. Sample Implementation (Python)

```
INF = float('inf')

def floyd_warshall(graph):
    n = len(graph)
    dist = [row[:] for row in graph]      # copy of adjacency matrix

    for k in range(n):
        for i in range(n):
            for j in range(n):
                if dist[i][k] + dist[k][j] < dist[i][j]:
                    dist[i][j] = dist[i][k] + dist[k][j]
    return dist


routers = ['R0', 'R1', 'R2', 'R3']
graph = [
    [0,   5, INF, 10],
    [INF, 0,   3, INF],
    [INF, INF, 0,   1],
    [INF, INF, INF, 0],
]

result = floyd_warshall(graph)
for i, row in enumerate(result):
    print(routers[i], row)
```

5. Sample Input and Output

Input: direct links R0→R1 = 5, R0→R3 = 10, R1→R2 = 3, R2→R3 = 1 (all other pairs unreachable directly).

```
R0 [0, 5, 8, 9]
R1 [inf, 0, 3, 4]
R2 [inf, inf, 0, 1]
R3 [inf, inf, inf, 0]
```

Validation: R0→R3 direct cost is 10, but routing through intermediates R1 and R2 (R0→R1→R2→R3 = 5+3+1 = 9) is cheaper, exactly as computed. This confirms the intermediate-node relaxation is working correctly.

6. Suggested Tools / Technologies

- Python (NetworkX) or C++ (Boost Graph Library) for offline route-table pre-computation.
- SDN controllers such as OpenDaylight or ONOS, which can push the computed shortest-path matrix directly into forwarding tables via OpenFlow.
- Graph databases (Neo4j) for visualizing and querying the router topology and path costs interactively.

7. Performance Analysis

Time Complexity: O(n^3) for n routers, since three nested loops each run n times. For n = 500 routers this is 1.25 × 10^8 operations — easily precomputed offline within seconds on modern hardware.

Space Complexity: O(n^2) to store the distance matrix (and optionally another O(n^2) next-hop matrix for path reconstruction).

Industry interpretation: O(n^3) is acceptable for backbone/core networks where the router count is in the hundreds and topology changes infrequently (recomputed on link-up/link-down events, not per packet). The resulting matrix lets edge routers do O(1) next-hop lookups, minimizing latency and improving overall network efficiency, though for very large (n > few thousand) or highly dynamic networks, repeated single-source Dijkstra runs or incremental shortest-path updates are usually preferred over full recomputation.

Question 3 — Huffman Coding: Data Compression

A video streaming platform compresses data based on symbol frequencies to optimize storage and transmission. Analyze how Huffman coding can be used to generate optimal prefix codes. Design the solution by explaining tree construction and encoding process. Suggest tools for implementing compression. Analyze compression efficiency and time complexity. Interpret the results in terms of bandwidth and storage optimization.

1. Understanding the Industry Problem

Video/audio streams, once quantized, are ultimately encoded as symbols (e.g., DCT coefficients, motion-vector residuals) with highly skewed frequency distributions — a few symbols occur very often, many occur rarely. Fixed-length codes waste bits on frequent symbols. The platform needs a lossless, variable-length prefix code that minimizes the expected number of bits per symbol without any ambiguity when decoding a bitstream.

- Key issue 1 — Prefix-free requirement: no code word may be a prefix of another, or the decoder cannot uniquely parse the bitstream.
- Key issue 2 — Frequency skew: symbol frequencies vary widely across a video segment, so a static fixed-length code is bandwidth-inefficient.
- Key issue 3 — Encoding/decoding must be fast enough for real-time streaming pipelines.

2. Why Huffman's Greedy Approach

Huffman coding builds an optimal prefix code by repeatedly merging the two least-frequent nodes into a new parent node whose frequency is their sum, and pushing this parent back into the candidate pool. This is a proven greedy-choice-property algorithm: at every step, merging the two globally least-frequent symbols never prevents an optimal solution from being reached (this is formally proved via an exchange argument), so no backtracking or dynamic-programming table is required — the greedy choice is always safe.

3. Solution Design — Tree Construction and Encoding

- Step 1: Create a leaf node for every symbol, weighted by its frequency, and insert all leaves into a min-priority-queue (min-heap).
- Step 2: Repeatedly extract the two nodes with the smallest frequency, create a new internal node with these two as children and frequency = sum of the two, and insert the new node back into the heap.
- Step 3: Repeat until exactly one node (the root) remains — this is the Huffman tree.
- Step 4: Traverse the tree from root to each leaf, appending '0' for a left branch and '1' for a right branch, to obtain each symbol's prefix code.

4. Sample Implementation (Python)

```
import heapq

class Node:
    def __init__(self, freq, symbol=None, left=None, right=None):
        self.freq, self.symbol, self.left, self.right = freq, symbol, left, right
    def __lt__(self, other):
        return self.freq < other.freq

def build_huffman(freq_table):
    heap = [Node(f, s) for s, f in freq_table.items()]
    heapq.heapify(heap)

    while len(heap) > 1:
        left = heapq.heappop(heap)
        right = heapq.heappop(heap)
        merged = Node(left.freq + right.freq, left=left, right=right)
        heapq.heappush(heap, merged)
    return heap[0]                    # root of the tree

def generate_codes(node, prefix='', codes=None):
    if codes is None:
        codes = {}
    if node.symbol is not None:         # leaf
        codes[node.symbol] = prefix or '0'
    else:
        generate_codes(node.left, prefix + '0', codes)
        generate_codes(node.right, prefix + '1', codes)
    return codes


freq_table = {'A': 5, 'B': 9, 'C': 12, 'D': 13, 'E': 16, 'F': 45}
root = build_huffman(freq_table)
codes = generate_codes(root)
for symbol in sorted(codes):
    print(symbol, ':', codes[symbol])
```

5. Sample Input and Output

Input frequency table: A=5, B=9, C=12, D=13, E=16, F=45 (total 100 symbol occurrences).

A : 0010
B : 0011
C : 000
D : 010
E : 011
F : 1

Validation: no code word is a prefix of another (prefix-free property holds), and the most frequent symbol F (45 occurrences) gets the shortest code (1 bit) while the least frequent symbol A (5 occurrences) gets a 4-bit code — exactly the expected inverse relationship between frequency and code length.

6. Suggested Tools for Implementation

- Python (heapq) or C/C++ (STL priority_queue) for building the tree; C/C++ for the production video codec path (speed-critical).
- FFmpeg / x264 / x265 internally use Huffman-style entropy coding (and its arithmetic-coding variant CABAC) — useful as a reference implementation.
- zlib/DEFLATE and Brotli libraries for general-purpose Huffman-based compression of metadata and manifest files alongside the video stream.

7. Performance Analysis

Time Complexity: O(n log n) for n distinct symbols — each of the (n-1) heap merge operations costs O(log n) for heap insert/extract.

Space Complexity: O(n) for the heap and the resulting binary tree (2n-1 nodes total).

Compression efficiency: for the sample table, the average code length is (5×4 + 9×4 + 12×3 + 13×3 + 16×3 + 45×1)/100 = 2.24 bits/symbol versus 3 bits/symbol for a fixed-length code over 6 symbols — roughly a 25% reduction in encoded size.

Industry interpretation: because Huffman coding is provably optimal among prefix codes for a known static frequency distribution, video platforms use it (or its arithmetic/range-coding refinements) as a final entropy-coding stage after prediction and quantization, directly cutting bandwidth cost and buffering delay for end users, especially over constrained mobile networks.

Question 4 — Minimum Spanning Tree: Network Design

An ISP wants to connect multiple cities with minimum cable cost. Analyze how Prim's or Kruskal's algorithm can be applied to construct a Minimum Spanning Tree. Design a feasible solution by comparing both algorithms. Suggest appropriate tools and data structures for implementation. Analyze the time complexity of each approach. Interpret the results in terms of cost optimization and practical network design.

1. Understanding the Industry Problem

The ISP must lay physical fiber/cable to connect every city in its service region so that all cities are reachable, while minimizing the total cost of cable laid. Modeling cities as vertices and candidate cable routes (with their laying cost) as weighted edges, the requirement — connect all vertices with the least total edge weight and no cycles — is exactly the Minimum Spanning Tree (MST) problem.

- Key issue 1 — Every city must be connected (spanning), but redundant/cyclic links waste capital cost.
- Key issue 2 — Multiple candidate routes between city pairs exist with different costs (terrain, distance, permits).
- Key issue 3 — Network must remain a tree (n-1 edges for n cities) to minimize cost, though real deployments may add limited redundancy afterward for fault tolerance.

2. Comparing Kruskal's and Prim's Algorithms

Kruskal's algorithm (edge-based, greedy): sort all edges by cost ascending; repeatedly add the cheapest edge that does not form a cycle (checked via a Union-Find/Disjoint-Set structure), until n-1 edges are chosen.

Prim's algorithm (vertex-based, greedy): start from an arbitrary city and grow a single tree by repeatedly adding the cheapest edge that connects a city already in the tree to a city outside it (typically using a min-priority-queue).

- Both algorithms rely on the Cut Property (the cheapest edge crossing any cut of the graph is safe to include in some MST) and both are greedy-optimal — they always produce a true minimum spanning tree.
- Kruskal's is preferable for sparse networks (few candidate cable routes relative to cities) because it processes edges directly; Prim's is preferable for dense networks (many candidate routes) because it avoids sorting all edges upfront and works naturally with an adjacency-matrix/heap representation.

3. Solution Design (Kruskal's)

1. Represent each city as an element of a Union-Find (Disjoint Set) structure.
2. Sort all candidate cable edges by cost in ascending order.
3. For each edge (u, v, cost), in sorted order:
   if find(u) != find(v): # adding it will not create a cycle
      union(u, v)
      add edge to MST
4. Stop when the MST contains (n - 1) edges.

4. Sample Implementation (Python — Kruskal's with Union-Find)

```
class DisjointSet:
    def __init__(self, nodes):
        self.parent = {n: n for n in nodes}
    def find(self, x):
        while self.parent[x] != x:
            self.parent[x] = self.parent[self.parent[x]]   # path compression
            x = self.parent[x]
        return x
    def union(self, x, y):
        rx, ry = self.find(x), self.find(y)
        if rx == ry:
            return False
        self.parent[rx] = ry
        return True

def kruskal_mst(cities, edges):
    ds = DisjointSet(cities)
    mst, total_cost = [], 0
    for u, v, cost in sorted(edges, key=lambda e: e[2]):
        if ds.union(u, v):
            mst.append((u, v, cost))
            total_cost += cost
    return mst, total_cost


cities = ['A', 'B', 'C', 'D', 'E']
edges = [
    ('A', 'B', 4), ('A', 'C', 3), ('B', 'C', 2),
    ('B', 'D', 5), ('C', 'D', 6), ('C', 'E', 7), ('D', 'E', 1),
]
```

```
mst, total_cost = kruskal_mst(cities, edges)
print('Cables to lay:', mst)
print('Total minimum cable cost:', total_cost)
```

5. Sample Input and Output

Input: 5 cities A–E, candidate cable routes with costs as listed in the code above.

Cables to lay: [('D', 'E', 1), ('B', 'C', 2), ('A', 'C', 3), ('B', 'D', 5)]
Total minimum cable cost: 11
Validation: 4 edges are selected for 5 cities (n-1 = 4, correct for a spanning tree); the edge A-B (cost 4) was correctly skipped because A, B, C were already connected via A-C and B-C, and including A-B would have formed a cycle without reducing cost.

6. Suggested Tools and Data Structures

- Union-Find (Disjoint Set Union) with path compression and union by rank — essential for Kruskal's cycle detection in near O(1) amortized time.
- Min-priority-queue (binary heap / Fibonacci heap) — essential for Prim's edge selection.
- Python NetworkX (minimum_spanning_tree), GIS tools such as QGIS/ArcGIS for overlaying the MST on real geographic city coordinates, and network-planning suites for fiber-route cost estimation.

7. Performance Analysis

Kruskal's Time Complexity: O(E log E) dominated by sorting the E candidate edges, plus near-O(E) for Union-Find operations with path compression and union by rank.

Prim's Time Complexity: O(E log V) using a binary heap, or O(E + V log V) using a Fibonacci heap.

Space Complexity: O(V + E) for both — edge list / adjacency list plus the auxiliary Union-Find or heap structure.

Industry interpretation: for a nationwide ISP rollout where the number of candidate routes (E) is close to the number of cities (V) — i.e., a sparse candidate graph — Kruskal's O(E log E) is typically faster and simpler to implement; for a metro deployment where nearly every city pair is a viable route (dense graph), Prim's O(E + V log V) with a Fibonacci heap scales better. In both cases the MST guarantees the lowest total capital expenditure on cable while still connecting every city, directly translating into cost savings for the ISP's network build-out.

Question 5 — Job Sequencing with Deadlines: Profit Maximization

A company must schedule jobs with deadlines and associated profits such that maximum profit is earned within given time constraints. Analyze how the Job Sequencing with Deadlines problem can be solved using a Greedy approach. Design a solution by explaining the selection and scheduling strategy. Justify why the greedy choice leads to optimal profit. Suggest appropriate tools and data structures for implementation. Analyze time complexity and interpret results in terms of business profit optimization.

1. Understanding the Industry Problem

Each job has a deadline (the latest time slot by which it must complete) and a profit (earned only if it completes by its deadline). Every job takes exactly one unit of time, and only one job can run in any time slot. The company wants to choose and schedule a subset of jobs — each into some slot at or before its deadline — that maximizes total profit.

- Key issue 1 — Slot conflicts: two jobs cannot occupy the same time slot, so scheduling order matters.
- Key issue 2 — Deadline feasibility: a job scheduled after its deadline earns nothing, so late slots must be filled carefully.
- Key issue 3 — Not all jobs can be accepted if the number of jobs exceeds the maximum deadline (number of available slots).

2. Why the Greedy Approach Is Optimal

The greedy strategy is: sort jobs by profit in descending order, and for each job (highest profit first) try to place it in the latest available time slot at or before its deadline (scanning backward from its deadline down to slot 1). This is optimal because of an exchange-argument proof: if a higher-profit job is left unscheduled while a lower-profit job occupies a slot the higher-profit job could have used, swapping them can only increase (or keep equal) total profit — so any optimal schedule can always be transformed into the one the greedy algorithm produces without loss. Placing each accepted job as late as possible (rather than as early as possible) is what keeps earlier slots free for other jobs with tighter deadlines, which is the crux of correctness.

3. Solution Design

1. Sort all jobs in descending order of profit.
2. Let max_deadline = maximum deadline among all jobs; create max_deadline empty slots.
3. For each job (in profit-descending order):
    scan slots from min(job.deadline, max_deadline) down to 1
    if an empty slot is found:
        place the job there; add its profit to the total
        stop scanning for this job

4. Jobs that found no free slot are rejected (left unscheduled).

4. Sample Implementation (Python)

```
def job_sequencing(jobs):
    # jobs: list of (job_id, deadline, profit)
    jobs = sorted(jobs, key=lambda j: j[2], reverse=True)   # sort by profit desc
    max_deadline = max(j[1] for j in jobs)
    slots = [None] * (max_deadline + 1)                    # 1-indexed, slot[0] unused

    total_profit, schedule = 0, [None] * (max_deadline + 1)
    for job_id, deadline, profit in jobs:
        for t in range(min(deadline, max_deadline), 0, -1):
            if schedule[t] is None:
                schedule[t] = job_id
                total_profit += profit
                break

    sequence = [schedule[t] for t in range(1, max_deadline + 1) if schedule[t]]
    return sequence, total_profit


jobs = [
    ('J1', 2, 100),
    ('J2', 1, 19),
    ('J3', 2, 27),
    ('J4', 1, 25),
    ('J5', 3, 15),
]

sequence, total_profit = job_sequencing(jobs)
print('Optimal job sequence (slot order):', sequence)
print('Maximum total profit:', total_profit)
```

5. Sample Input and Output

Input: J1(deadline=2, profit=100), J2(deadline=1, profit=19), J3(deadline=2, profit=27), J4(deadline=1, profit=25), J5(deadline=3, profit=15).

Optimal job sequence (slot order): ['J3', 'J1', 'J5']
Maximum total profit: 142
Validation: Slot1=J3, Slot2=J1, Slot3=J5. J2 (profit 19) and J4 (profit 25) are correctly rejected because, after J1 and J3 (both deadline 2, higher profit) occupy slots 1 and 2, no earlier slot

remains for J2/J4's deadline-1 requirement — a smaller total profit than sacrificing J1 or J3 would result, confirming the greedy choice is optimal.

6. Suggested Tools and Data Structures

- An array/list of boolean slot markers (as above) for small deadline ranges, or a Disjoint-Set 'next free slot' structure for large deadline ranges to speed up slot search.
- A max-heap (priority queue) if jobs arrive as a stream rather than a static list.
- Enterprise scheduling engines (e.g., IBM ILOG CPLEX, OptaPlanner) for extending this to multi-resource, multi-slot-per-job real business scheduling problems.

7. Performance Analysis

Time Complexity: O(n log n) for sorting jobs by profit, plus O(n × d) for the slot-search step in the naive array version (n jobs, d = max deadline); using a Disjoint-Set 'find next free slot' optimization reduces this to O(n α(n)) ≈ O(n) after sorting, i.e. overall O(n log n).

Space Complexity: O(d) for the slot array (or O(n) for the Disjoint-Set version), plus O(n) to hold the job list.

Industry interpretation: because the algorithm is near-linear after an initial sort, it scales comfortably to thousands of jobs (e.g., ad-slot allocation, machine-time booking, delivery-slot assignment), letting a business re-run the optimization whenever new high-value jobs arrive and immediately see the maximum achievable profit under the current deadline constraints — directly supporting revenue-maximizing operational decisions.

RUBRICS FOR CALCULATION

Industry Problem Solving Task

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%				
Application of Engineering Concepts	25%				
Solution Design	25%				
Use of Tools	15%				
Analysis	10%				
Professionalism	5%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:

Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____